

Chicago Crime Composition Analysis (2001-2026)

Compositional structural-break analysis of monthly reported crime, with COVID-era assessment

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Executive Summary

This analysis tests whether Chicago's crime composition underwent discrete regime shifts during the COVID-19 pandemic, using compositional data methods on 25 years of monthly reported-incident data. The central finding is that **Chicago's crime composition evolved gradually rather than in discrete regimes, and the data do not support COVID-19 as a structural break point**. Both confirmatory and exploratory multivariate changepoint detection (Dynp and Pelt on the centered log-ratio transformed series) place detected breaks before COVID or fail to identify any breaks at all under conventional evidence thresholds. The pandemic period coincides with the tail of a longer compositional drift that began before 2010 and continued through 2020.

A secondary finding concerns the per-era dependence structure between crime categories: while statistical correlations shifted measurably across era boundaries, the largest shifts trace to two declining high-volume categories (Drug Abuse Violations, Liquor Laws) releasing their compositional pressure on the rest of the system, rather than new co-movement clusters forming among other categories.

Data and Scope

The analysis uses monthly counts of reported crime incidents from the Chicago Data Portal, covering January 2001 through April 2026 (304 months). Crime types are aggregated to the 25 FBI Uniform Crime Reporting categories that achieved at least 90.8% monthly coverage. Involuntary Manslaughter / Reckless Homicide was excluded as too sparse for compositional modeling (23.4% coverage).

Era Definitions (administrative reference windows)

Era	Period	Months
Pre-COVID	January 2001 - February 2020	230
COVID	March 2020 - December 2022	34
Post-COVID	January 2023 - April 2026	40

Era boundaries are policy-defined: anchored to the WHO pandemic declaration (March 2020) and the expiration of Illinois's disaster proclamation (late 2022). The analysis treats these as administrative windows for descriptive comparison, not as empirically validated regime

boundaries. Section 6c of the notebook documents that data-driven changepoint detection does not endorse these specific dates.

Methodology

Counts are aggregated to a monthly composition (each category's share of total monthly incidents) and transformed via the Centered Log-Ratio (CLR) function. This places the data in the Aitchison simplex (a mathematical space designed specifically for analyzing **compositional data**), where standard multivariate methods apply while preserving compositional constraints.

Pseudocount selection

The CLR transformation requires a pseudocount ϵ to handle zero entries. $\epsilon = 0.00894$ was selected via grid search on stability and rank-preservation criteria. Sensitivity verified at the adjacent grid value ($\epsilon = 0.00812$): cosine similarity ≈ 1.0 between PCA decompositions. The analysis is stable in the neighborhood of the chosen ϵ .

Three-way sensitivity structure

Following Aitchison's framework, the primary analysis uses unscaled 25-category CLR (Version A). Two sensitivity specifications are reported throughout: z-scored CLR (Version B) tests robustness to magnitude dominance; vice-excluded 23-category CLR with recomputed geometric mean (Version C) tests robustness to two highly variable categories (Gambling, Prostitution). This A/B/C structure parallels Aitchison's recommendation when one or two categories dominate the variance.

Statistical methods

- Principal Component Analysis (pooled across the full 304-month dataset, with epsilon stability diagnostics)
- Per-era moment estimation with Ledoit-Wolf shrinkage on covariance matrices
- Stationarity testing: ADF, KPSS, and Zivot-Andrews with FDR correction across 24 dense categories
- Block bootstrap (block size 6 months, 10,000 iterations) for era-pair mean comparisons, with FDR correction
- Multivariate changepoint detection: Dynp confirmatory at $n=2$ breaks, Pelt exploratory with BIC penalty sweep across 5 multipliers and 4 min_size values
- Frobenius-distance comparison of per-era correlation matrices, with two within-era baselines (halves and short-window resampling)
- Element-wise correlation differences for identifying specific category-pairs driving cross-era shifts

Key Findings

Question 1: Do administrative pandemic-era boundaries correspond to data-driven structural breaks?

Answer: No. This is the most counterintuitive finding of the analysis.

Method	Version A (25 cats)	Version C (23 cats, no vice)	Outcome
Dynp (forced $n=2$)	2013-11, 2019-09	2008-06, 2015-12	Both pre-COVID
Pelt @ BIC penalty	0 breaks	0 breaks	No support at BIC
Pelt @ $0.25\times$ BIC	1 break: 2015-07	1 break: 2015-02	Single weak break, mid-2015
Pelt min_size sweep	0 breaks (12–36 mo)	—	Null is robust

- **Confirmatory Dynp (forced $n=2$ breaks):** Version A returned breaks at 2013-11 and 2019-09; Version C returned 2008-06 and 2015-12. Version A's later break (2019-09) sits 5 months before the assumed COVID-era start (2020-02), while its earlier break (2013-11) precedes the assumed COVID start by 75 months. Version C's two breaks are 84 and 140 months from the assumed era boundaries. None of the four detected break dates land within an assumed COVID-era boundary.
- **Exploratory Pelt with BIC penalty:** Returned zero breakpoints in both Version A and Version C. Across penalty multipliers from $0.5\times$ to $4\times$ BIC, no break clears the evidence threshold. Across minimum-segment-length choices from 12 to 36 months, the null holds.
- **Permissive penalty ($0.25\times$ BIC):** A single weak break emerges in mid-2015 in both versions. This coincides temporally with the Liquor Laws Zivot-Andrews break (2015-10) and vice category transitions, but does not survive BIC-level evidence.

Interpretation: the composition evolved gradually rather than in discrete regimes. The pandemic period coincides with the tail of a longer compositional drift, not a structural break point.

Univariate Zivot-Andrews testing corroborates the gradual-drift reading; only 3 of 24 dense categories showed statistically significant single structural breaks (Arson 2020-01, Prostitution 2020-09, Liquor Laws 2015-10), with Liquor Laws breaking five years before COVID.

Question 2: Which crime categories co-move within the data?

Pooled co-movement structure

PCA on the unscaled 25-category CLR matrix produces a dominant axis (PC1, 65.3% of variance). Removing the two vice categories (Gambling and Prostitution) reduces PC1 to 47.1% of variance and an 18-point reduction, indicating that the two vice categories contribute roughly

a third of the leading axis, but a strong co-movement structure persists among the remaining 23 categories. Z-scored CLR (Version B) reduces PC1 only to 56.7%, confirming the co-movement is genuine rather than a magnitude artifact.

Per-era dependence structure (Frobenius distance between correlation matrices)

All three cross-era Frobenius distances substantially exceed the 95th percentile of the short-window baseline (5.10), indicating that the dependence structure between crime categories shifted across each era boundary, not just the mean shares:

Comparison	Frobenius distance	vs. halves baseline	vs. short-window baseline
Pre-COVID ↔ COVID	8.11	0.94×	1.99× ★
COVID ↔ Post-COVID	5.74	0.66×	1.41× ★
Pre-COVID ↔ Post-COVID	9.14	1.06×	2.24× ★
Within Pre-COVID (baseline)	4.08 (mean) / 5.10 (95th)	—	1.00×

The Pre ↔ Post-COVID distance (9.14) is the largest, exceeding both Pre ↔ COVID (8.11) and COVID ↔ Post-COVID (5.74). The post-COVID dependence structure did not revert to the pre-pandemic pattern; the system's co-movement structure continued to evolve in the same direction after the pandemic subsided.

Specific category-pair shifts (element-wise correlation differences)

The 10 largest absolute correlation shifts between Pre-COVID and Post-COVID concentrate on two categories: Drug Abuse Violations (appearing in 5 of the top 10 shifts) and Liquor Laws (appearing in 5 of the top 10). Both show the same pattern: strong negative correlations with most other categories in Pre-COVID, swinging to near-zero or positive correlations in Post-COVID. Examples include Drug Abuse Violations ↔ Fraud (−0.59 → +0.44), Drug Abuse Violations ↔ Aggravated Assault (−0.69 → +0.18), and Liquor Laws ↔ Fraud (−0.78 → +0.35).

The math of shares, not behavior change

This pattern reflects the math of how shares are linked rather than a substantive realignment of crime types. In CLR space, when a single category occupies a large share of the composition, its month-to-month share movements must push every other category's share in the opposite direction, producing systemic negative correlations. Drug Abuse Violations had a pre-COVID mean of CLR ≈ 2.13 , among the largest mean shares in the data. Its sharp decline (Pre → Post shift of −1.03 in CLR units, the largest non-vice decline in the dataset) reduced its dominance of the composition. The negative correlations it imposed on other categories weakened or

reversed, which is a direct result of how shares are mathematically linked, not a behavioral shift in the other categories.

Bootstrap era comparisons (supporting evidence)

- **Pre-COVID → COVID:** All 24 dense categories show significant mean differences. The asymmetric window lengths (230 vs 34 months) mean that part of this signal reflects long-running drift rather than a discrete shock.
- **COVID → Post-COVID:** Comparing two windows of similar length, only 3 of 24 categories remain significant after FDR correction (Homicide, Weapons Violations, Burglary). The system largely stabilized rather than continuing to reorganize.
- **Pre-COVID → Post-COVID:** 21 of 24 categories significantly different. Compositional changes persisted rather than fully reverting.

Conclusions

Chicago's crime composition evolved gradually over the 2001-2026 dataset. The administrative pandemic-era boundaries do not align with empirical structural breaks; the composition's changes accumulated over decades rather than concentrating at policy boundaries. The pandemic period coincides with the tail of a longer compositional shift that began years earlier and continued afterward.

The dependence structure between crime categories also shifted across era boundaries, but the largest shifts reflect how shares are mathematically linked to the decline of previously-dominant categories, releasing their pressure on other shares rather than the emergence of new co-movement clusters.

These findings are consistent across multiple analytical methods: (1) most crime categories are non-stationary across the dataset (21 of 24 dense categories); (2) only three categories show statistically significant single structural breaks under Zivot-Andrews testing (Arson, Prostitution, Liquor Laws), with Liquor Laws breaking in 2015, five years before COVID; (3) PC1 trajectory shows a continuous descent through the dataset without abrupt regime changes; (4) multivariate changepoint detection finds no breaks at the assumed era boundaries under conventional BIC penalties (Version A's nearest detected break, 2019-09, is 5 months before the assumed COVID start but does not survive Pelt's evidence threshold); (5) the post-COVID dependence structure is further from pre-COVID than COVID itself was, indicating continued evolution rather than reversion.

Limitations & Methodological Notes

- **Reporting lag.** The Chicago Data Portal backfills recent months for several weeks. The post-2025 data is partly affected by this lag; interpretation of the post-COVID tail should account for the possibility that the most recent months are undercounted.
- **Enforcement vs occurrence.** Reported incident counts measure police activity, not crime occurrence. For enforcement-driven categories (Drug Abuse Violations, Liquor Laws, Gambling, Prostitution, and Weapons Violations), counts respond to enforcement priorities as much as to underlying behavior. Because CLR shares are linked by definition (when one share changes, all others must change), shifts in enforcement of high-volume categories propagate to all other categories' share statistics. This caveat conditions all CLR-based findings, including the dependence-structure shifts attributed to declining vice and drug categories.
- **Per-era PCA was not attempted.** The COVID era (34 months) and Post-COVID era (40 months) provide an insufficient sample size for stable eigenvector estimation at 25 dimensions. The Frobenius and element-wise correlation analyses answer the per-era co-movement question at the level supported by the sample sizes, without committing to specific axis rotations that the data cannot reliably estimate.
- **Compositional analysis is share-based.** All findings describe changes in the relative composition of crime types, not changes in absolute crime levels. A category declining in

CLR share is consistent with either a true decline in incidents or a relative shrinkage caused by other categories growing.

- **Eras are administratively defined.** The Pre-COVID / COVID / Post-COVID windows are anchored to policy events, not data-driven regime boundaries. Section 6c of the notebook documents that data-driven changepoint detection does not endorse these specific dates; the eras are retained for descriptive comparison with prior work and policy discourse.

Technical Summary

- **Observations:** 304 months. The raw 26-category dataset covered 7,904 expected cells at 96.5% completeness (277 gaps). After excluding Involuntary Manslaughter / Reckless Homicide (23.4% coverage), the analytical 25-category dataset covered 7,600 cells at 99.4% completeness; the 44 remaining gaps were filled with zeros, with audit flags retained for traceability.
- **CLR pseudocount:** $\epsilon = 0.00894$ (selected via grid sweep on stability and rank-preservation criteria; verified stable at the adjacent grid value $\epsilon = 0.00812$)
- **PCA variance explained:** PC1 = 65.3% (Version A, unscaled 25-cat) / 56.7% (Version B, z-scored 25-cat) / 47.1% (Version C, unscaled 23-cat, vice-excluded)
- **Significant Zivot-Andrews breaks (FDR-adjusted, $\alpha=0.05$):** Arson 2020-01, Prostitution 2020-09, Liquor Laws 2015-10 — 3 of 24 dense categories
- **Dynp confirmatory breaks ($n_bkps=2$):** Version A: 2013-11, 2019-09 | Version C: 2008-06, 2015-12
- **Pelt breaks at BIC penalty and above:** Zero, in both Version A and Version C, across `min_size` values 12–36 months
- **Cross-era Frobenius distances:** Pre $\leftrightarrow$ COVID = 8.11, COVID $\leftrightarrow$ Post = 5.74, Pre $\leftrightarrow$ Post = 9.14 (all exceed 95th-percentile baseline of 5.10)
- **Largest dependence-structure drivers:** Drug Abuse Violations and Liquor Laws (each appearing in 5 of the top 10 Pre $\rightarrow$ Post correlation shifts)

Companion notebook: `ChicagoCrimeDetection.ipynb`. All numerical results in this document trace directly to specific code cells in the notebook.